

Dimitris Deliakidis

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Education

University of Pennsylvania

Expected May 2028

B.S.E. in Electrical Engineering; Accelerated M.S.E. in Electrical Engineering

Minor: Computer Science, Benjamin Franklin Scholar

Relevant Coursework: Embedded Systems, Circuit Design, Computer Architecture, Algorithms

Experience

Undergraduate Research Assistant

Summer 2026

RWTH Aachen University — Acoustic Droplet Bioprinting, Aachen, Germany

- Built the computer-vision and instrumentation toolchain for an acoustic droplet-ejection bioprinter; developed a classical OpenCV pipeline that detects and indexes printed 6×6 droplet grids and automatically scores count, spacing, size uniformity, and satellite artifacts.
- Developed an automated bioprinter-to-laptop transfer pipeline, offline OPBOX ultrasound peak-logging tools, and Arduino MAX6675 thermocouple firmware for live bio-ink temperature monitoring.

Research Assistant — ModLab, GRASP Laboratory

Jan 2026 – Present

GRASP Laboratories

- Built an improved LED-based 6-axis force/torque sensor from the ground up, combining 2 custom PCB stacks, a compliant PDMS structure, and 24-channel optical sensing in a fingertip-scale $27 \text{ mm} \times 20.8 \text{ mm}$ form factor.
- Improved measurement sensitivity by 35% through optical redesign, noise reduction, and ML-based calibration using 20/24 denoised channels at 500 Hz.

Subsystem Hardware Engineer Lead

Sep 2024 – Present

Penn Electric Racing Formula SAE EV

- Leading an 8-member engineering team in the development and electrical design of the car's Dashboard and Data Acquisition (DAQ) power system.
- Designed in Altium and simulated the primary power supply in LTspice, integrating the team's first supercapacitor-based backup circuit enabling safe MCU shutdown during power loss; improved efficiency by 15% (up to 95%) and reduced board space usage by 20%.
- Designed the team's first hardware-in-the-loop testing board (STM32, CAN, I2C, current sensing), reducing board validation time by two months.

Teaching Assistant — Dept. of Electrical & Systems Engineering

Jan 2025 – Present

University of Pennsylvania

- TA for ESE 2150 (Electrical Circuits and Systems): supporting lectures and the weekly hands-on lab (LabVIEW, SPICE) covering circuit analysis, 1st/2nd-order circuits, transfer functions, diodes, and MOSFET amplifiers.
- TA for the introductory ESE course (80-student section); collaborated with the instructor to draft a course textbook and update course materials.

Projects

- **Predicting 30-Day Diabetic Readmission** (CIS 5450, Big Data Analytics): Built an end-to-end ML pipeline over $\sim 100\text{k}$ US hospital encounters (DuckDB/Polars wrangling, ICD regex grouping, KMeans acuity features); used a patient-level GroupShuffleSplit and in-fold SMOTE to compare Logistic Regression, Random Forest, and XGBoost (ROC AUC ≈ 0.62), then calibrated the decision threshold to the clinical cost trade-off.
- **The Cost of Stability: A Comparative Evaluation of 6T, 8T and 10T SRAM Cells Across Technology Nodes** (ESE 5760): Characterized 6T/8T/10T SRAM cells in SPICE across seven process nodes (180 nm–22 nm) over five figures-of-merit (HSNM/RSNM/WSNM, static N-curve SINM/SVNM, and read access time), quantifying the area–stability–performance trade-off as CMOS scales.
- **Whear — RFID Smart Closet** (ESE 3500): Built a bare-metal STM32F411RE closet-inventory system; wrote a from-scratch UHF RFID (Impinj R2000) driver, an IRQ-driven UART path, and bare-register SPI LCD and bit-banged WS2812 drivers, with an ESP32→Firestore uplink and SwiftUI iOS app at sub-3 s end-to-end latency.
- **Analog Metal Detector** (ESE 3190): Designed, simulated, laid out, and bench-tested an analog metal detector from discrete MOSFETs — two LC oscillators, a differential-pair mixer, cascaded common-source amplifiers, an RC filter, and a power source-follower driving a speaker; a ferrous target shifts the audio tone from $\sim 1.2 \text{ kHz}$ to $\sim 2.6 \text{ kHz}$, matching SPICE and hand calculations within a few percent.
- **Mesh Network for Hiking Trails**: Designed and deployed a 10-node wireless mesh network (XBee, GPS, STM32) enabling off-grid communication and real-time monitoring as distress-signal points along hiking trails; led end-to-end hardware, firmware, and field testing.

Skills

• Altium • LTspice / SPICE • Python • C/C++ • OpenCV • STM32 • ESP32 • SolidWorks • Fusion 360 • Java • IoT

Volunteering Experience

MOm / Hellenic Society for the Study & Protection of the Monk Seal

Jun – Jul 2025

Led the information center on the island of Alonissos. Handled logistics for the souvenir shop and developed a Python script to automate the weekly volunteer scheduling process.

Selected Awards & Honors

- **Patent** owner: “Device for measuring and wireless transmission of seawater or freshwater quality parameters”
- **Full-ride scholarship** covering tuition and expenses for an Electrical Engineering degree at the University of Pennsylvania
- European Union Contest for Young Scientists (**EUCYS**) finalist (Sept 2022)
- International Science and Engineering Fair (**ISEF**) finalist (May 2022)